Modeling and Validation of a Soccer Ball's Three-Dimensional Flight to the Goal

Problem B

Abstract

We compare gravity, drag, constant-force and Magnus-force models. M3 estimates constant acceleration from three positions; M4 fits drag and linear-spin lift parameters. The predicted M4 goal-plane crossing for the competition shot is $(y, z) = (2.616, 1.953)$ m.

Keywords: Soccer ball, aerodynamic drag, Magnus effect, inverse prediction, trajectory comparison

Contents

1 Introduction **3**
1.1 Problem Background . . . 3
1.2 Problem Restatement . . . 3

2 Preparation of the Models **4**
2.1 Assumptions . . . 4
2.2 Notations . . . 5

3 Model Establishment **5**
3.1 Geometry and Goal Plane . . . 6
3.2 Force Analysis . . . 6
3.2.1 Air Buoyancy . . . 6
3.2.2 Gravity . . . 7
3.2.3 Aerodynamic Drag . . . 7
3.2.4 Magnus Lift . . . 8
3.3 Dynamical Equations . . . 8

4 Model Solving **9**
4.1 Forward Prediction . . . 9
4.1.1 Prediction procedure and experimental conditions . . . 9
4.1.2 Calculated trajectories and physical interpretation . . . 10
4.2 Inverse Prediction . . . 12
4.2.1 Early-position fitting and representative solution . . . 12
4.2.2 Comparison with recorded match flights . . . 13
4.2.3 Competition Problem B . . . 15
4.3 Case Studies: Different Penalty-Kick Techniques . . . 16

5 Sensitivity Analysis **20**

6 Discussion of Other Factors **20**
6.1 Asymmetric Surface Geometry . . . 20
6.2 Rotation and Aerodynamic Damping . . . 22
6.3 Wind Speed and Force Estimates . . . 22

7 Conclusions **23**

8 Model Evaluation **23**
8.1 Physical Interpretation and Computational Use . . . 23
8.2 Predictive Performance and Improvements . . . 23

References **24**

Reproduction Code **26**

1 Introduction

1.1 Problem Background

A penalty kick places stringent spatial and temporal demands on the goalkeeper. The goalkeeper must reach an appropriate location at the right time, while deciding when and in which direction to move. Studies of penalty anticipation discuss this timing problem and show that shot speed and kicking technique affect the predictability of shot direction [1, 2]. Such work also considers cues from the kicker before contact; the present problem instead focuses on predicting the ball's later position from its measured positions after launch.

The path of a kicked soccer ball is not generally a straight line. Gravity bends the trajectory downwards, air resistance slows the ball, and rotation can produce a sideways or vertical aerodynamic force. Bray and Kerwin combined aerodynamic forces with equations of motion and compared the resulting trajectories with experimental free kicks [5]. Myers and Mitchell analyzed three-dimensional spinning-ball motion and the influence of drag and spin [15]; Maponi and Piombin developed a numerical framework that also accommodates wind [12].

The aerodynamic behavior is also affected by the ball's surface. Wind-tunnel measurements by Asai et al. examined the relationship between flow conditions and soccer-ball forces [16], while Paudel and Yan's simulations compared a football with realistic seam geometry against a smooth sphere [14]. These studies motivate a spherical geometric approximation together with an examination of how aerodynamic coefficients affect the predicted flight.

Measurements of real flights provide a basis for evaluating such models. Griffiths et al. tracked the three-dimensional motion and spin of a marked football in field conditions [3]. Goff and Carré used camera-recorded trajectories to estimate aerodynamic coefficients [20]; subsequent work by Goff et al. combined measurements from 29 flights to construct speed-dependent drag and lift curves [4].

In this problem, the initial position and two early positions are given, whereas the initial velocity and spin are not directly specified. We therefore use the equations of motion to find states compatible with the early observations and calculate their goal-plane crossings. The DFL open dataset combines match events and optical tracking data for professional matches [10, 11], allowing predicted crossings to be compared with later recorded positions.

1.2 Problem Restatement

We model a soccer ball travelling in three dimensions from the vicinity of the penalty mark to the goal. The goal is 7.32 m wide and 2.44 m high; the penalty mark is 11 m from the goal line. We establish a model incorporating gravity, buoyancy, drag and the Magnus effect, solve for its trajectory and the point at which it crosses the goal plane, and compare the calculated positions with a recorded penalty flight. The prediction uses the initial position and the positions at 0.05 and 0.10 s; subsequent positions are reserved for validation. Section 2 states the assumptions and notation. Section 3 establishes the forces and equations. Section 4 compares forward and inverse predictions. Section 5 is reserved for sensitivity analysis. Section 6 discusses surface geometry, spin damping and wind, followed by conclusions and model evaluation in Sections 7 and 8.

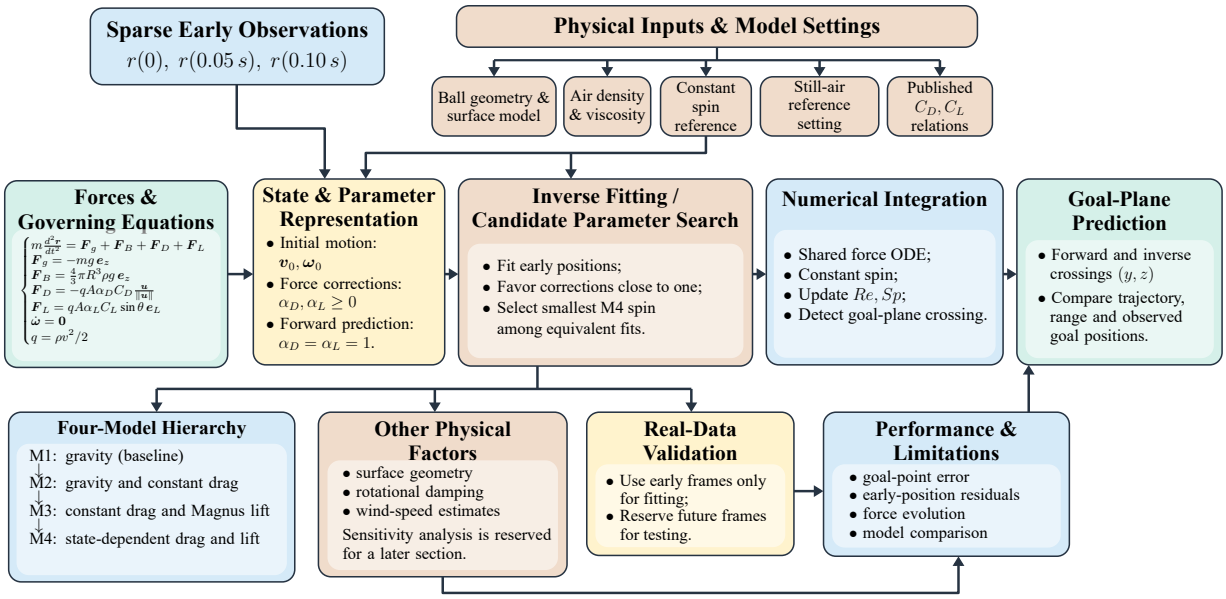


Figure 1: Force model including buoyancy, four-model baseline comparison and goal-plane validation.

2 Preparation of the Models

2.1 Assumptions

- **Ball–foot interaction.** Free flight begins at $t = 0$ immediately after the ball leaves the player's foot. The impact process and deformation during contact are not modeled.
- **Rigid spherical ball.** The ball is idealized as a rigid sphere with constant mass, radius and cross-sectional area $A = \pi R^2$; its centre of mass coincides with its geometric centre [12, 5, 15]. Empirical football coefficients represent the average aerodynamic effect of its surface. Section 6.1 discusses seams and asymmetric geometry.
- **Free-flight motion.** Gravity, air buoyancy, aerodynamic drag and spin-induced lift act during free flight [12, 5]. The flight considered ends at the first contact with the ground, goal structure or another object; collisions, bouncing and rolling are excluded. Integration also stops at the selected goal plane.
- **Uniform gravity.** Gravitational acceleration is constant, with $g = 9.81\,\mathrm{m/s^2}$.
- **Atmospheric properties.** Air density and dynamic viscosity are spatially uniform during each flight. Aerodynamic forces are represented using empirical drag and lift coefficients.
- **Wind.** The main calculation uses still air. When wind is included, its velocity vector is spatially and temporally constant during each flight; small-scale turbulence and rapidly varying gusts are neglected. Section 6.3 examines how different wind conditions affect relative airflow and force magnitude.

- **Spin during flight.** The main model neglects aerodynamic torque, so its spin vector remains constant. Section 6.2 examines this approximation by adding rotational damping to the same initial state.
- **Observations and unknown parameters.** The position coordinates supplied in the problem are treated as exact. The initial position and the next two early positions are the inverse input; subsequent DFL tracking positions are reserved for comparison [10, 11]. For bounded uncertainty analysis, unresolved initial velocity, spin, aerodynamic parameters and environmental conditions are constrained to stated physically reasonable ranges. Prediction uncertainty is attributed to these unresolved quantities and model assumptions rather than measurement noise. The recorded DFL vertical coordinate is interpreted as height in metres.

2.2 Notations

Table 1: Principal quantities and competition reference values.

Symbol	Meaning	Value or unit
$\boldsymbol{r}, \boldsymbol{v}$	Ball-centre position and velocity	m; m/s
$\boldsymbol{\omega}, \omega$	Spin vector and its magnitude	rad/s
m, R, A	Mass, radius and frontal area	0.43 kg; 0.11 m; πR^2
g	Gravitational acceleration	9.81 m/s^2
$\boldsymbol{u}, v$	Air-relative velocity and its magnitude	m/s
ρ	Air density	1.2 kg/m^3
$\nu, \mu = \rho\nu$	Kinematic and dynamic viscosity	1.50×10^{-5} m^2/s; Pa s
Re, Sp	Reynolds number and spin parameter	dimensionless
C_D, C_L	Drag and transverse lift coefficients	dimensionless
α_D, α_L	Multiplicative force corrections	dimensionless
$\theta, \boldsymbol{e}_L$	Spin–airflow angle and Magnus direction	rad; unit vector
X_g	Local goal-plane coordinate	11 m
$\boldsymbol{Q}, I_b$	Damping torque and moment of inertia (Section 6)	N m; kg m^2

The competition mass, radius and gravity follow the reference values of Maponi and Piombin [12]; the air density follows Asai et al. [16]. The viscosity is a declared reference air setting, with $\mu = \rho\nu$. Controlled experiments use their reported mass, diameter and air properties.

3 Model Establishment

The same Newtonian equations generate all four models by selecting the included aerodynamic terms. We first describe the geometry, then connect the forces to the surrounding flow and ball rotation.

3.1 Geometry and Goal Plane

For a pitch 105 m long with its origin at the centre, the two goal lines are located at $x = \pm 52.50$ m. The selected kick travels towards the positive-x goal. For the competition geometry the valid goal opening is $-3.66 \le y \le 3.66$ m and $0 \le z \le 2.44$ m. The goal-plane position is the ball centre's first forward intersection with $x = X_g$.

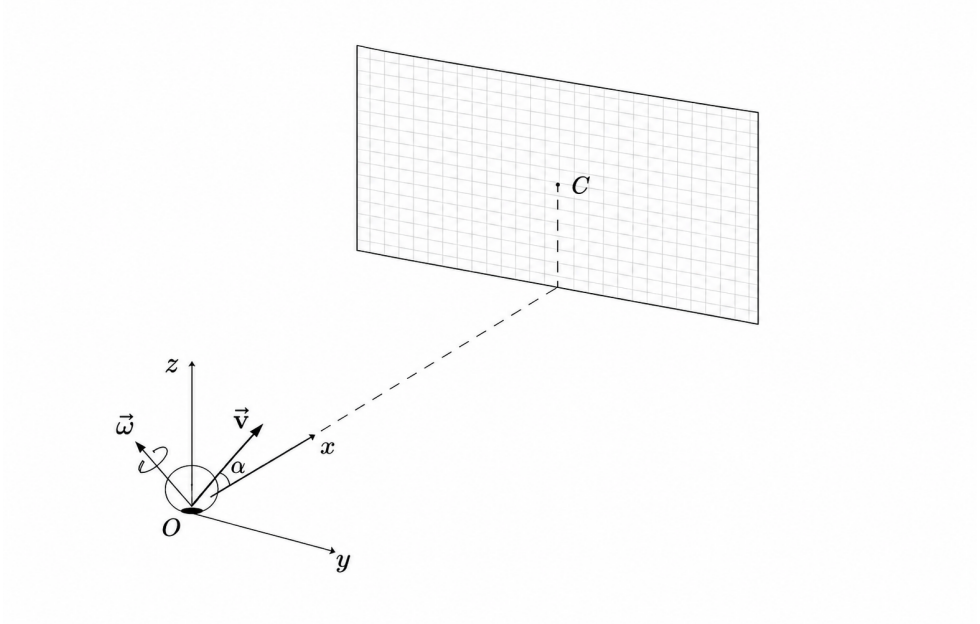


Figure 2: Penalty-kick geometry in local coordinates. O is the penalty mark and C is an illustrative point on the goal plane.

The competition coordinates use the penalty mark as the origin and place the goal plane at $x = 11$ m. For the match record, the pitch-centred coordinate system places the goal at $X_g = 52.50$ m.

The ball crosses the opening cleanly when its centre satisfies $|y| \le 3.66 - R$ and $R \le z \le 2.44 - R$ at $x = X_g$. Ground contact occurs when the descending centre reaches $z = R$. These event definitions are used consistently for all four models.

3.2 Force Analysis

3.2.1 Air Buoyancy

By Archimedes' principle the displaced air exerts an upward force on the ball. Its volume is $V_b = \frac{4}{3}\pi R^3$, giving

$$\boldsymbol{F}_B = \rho V_b g\, \boldsymbol{e}_z = \frac{4}{3}\pi R^3 \rho g\, \boldsymbol{e}_z. \tag{1}$$

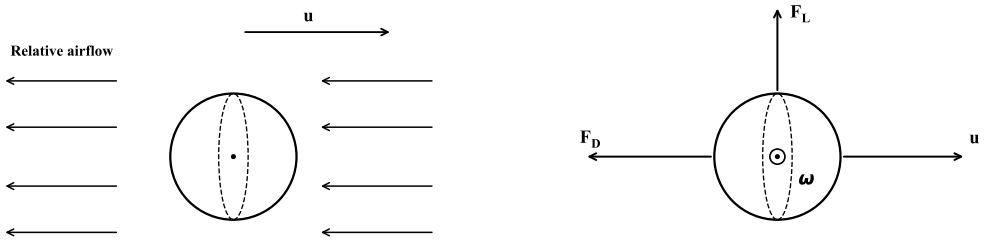
Together, gravity and buoyancy give $\boldsymbol{F}_g + \boldsymbol{F}_B = -(m - \rho V_b) g \boldsymbol{e}_z$. For the reference parameters in the table, buoyancy is about 1.56% of the ball's weight and changes the downward acceleration from g to $g(1 - \rho V_b/m)$.

3.2.2 Gravity

Gravity supplies a constant downward force,

$$\boldsymbol{F}_g = -mg\boldsymbol{e}_z, \tag{2}$$

where $\boldsymbol{e}_z$ is the upward vertical unit vector. It produces downward curvature even when all aerodynamic forces are removed.



(a) Airflow in the ball frame (b) Example: spin out of the page

Figure 3: Relative airflow and force directions. The symbol $\odot$ denotes spin out of the page; $\boldsymbol{u}$ in the left panel is the velocity of the ball relative to the air.

3.2.3 Aerodynamic Drag

Let $\boldsymbol{u} = \boldsymbol{v} - \boldsymbol{v}_{\text{wind}}$ be velocity relative to the air, $u = \|\boldsymbol{u}\|$ and $A = \pi R^2$. Drag acts opposite to relative velocity and scales with its square [20, 17]:

$$\boldsymbol{F}_D = -\tfrac{1}{2}\rho A C_D u \boldsymbol{u}. \tag{3}$$

The dimensionless drag coefficient C_D is constant in the main model.

3.2.4 Magnus Lift

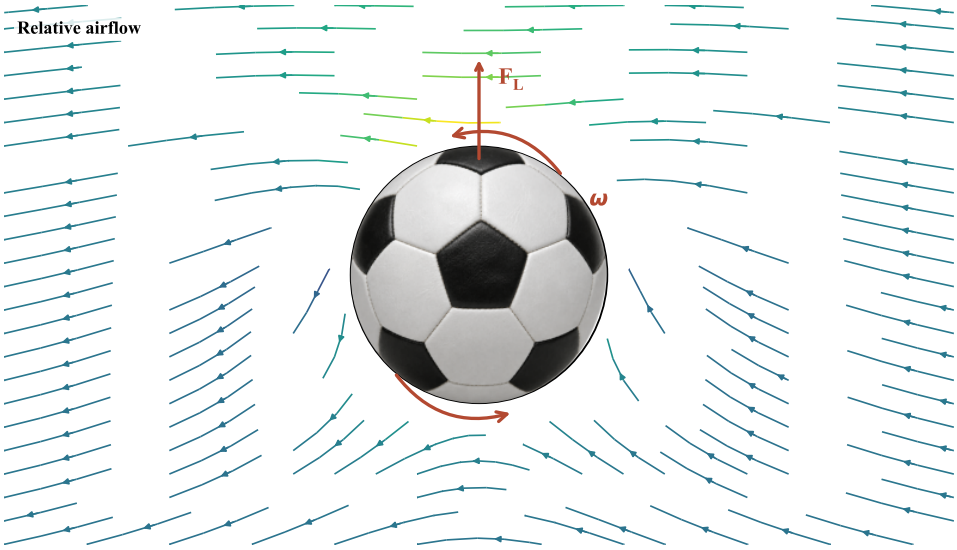


Figure 4: Idealized flow around a spinning football.

For relative speed u, let $Sp = R\|\boldsymbol{\omega}\|/u$ and use the simple relation $C_L = \alpha Sp$. Resolving the transverse component gives

$$\boldsymbol{F}_L = \tfrac{1}{2}\rho A R \alpha (\boldsymbol{\omega} \times \boldsymbol{u}). \tag{4}$$

The force is zero for zero spin or relative velocity. Reversing the cross-product order requires reversing the sign.

3.3 Dynamical Equations

Newton's second law and angular-momentum balance give

$$m\dot{\boldsymbol{v}} = \sum \boldsymbol{F}, \qquad \frac{d\boldsymbol{L}}{dt} = \boldsymbol{Q}, \qquad \boldsymbol{L} = I_b \boldsymbol{\omega}. \tag{5}$$

With zero applied torque and constant spherical inertia, the main model is

$$\begin{cases} \dot{\boldsymbol{r}} = \boldsymbol{v}, \\ m\dot{\boldsymbol{v}} = \boldsymbol{F}_g + \boldsymbol{F}_B + \boldsymbol{F}_D + \boldsymbol{F}_L, \\ \dot{\boldsymbol{\omega}} = \boldsymbol{0}. \end{cases} \tag{6}$$

Its initial conditions are

$$\begin{cases} \boldsymbol{r}(0) = \boldsymbol{r}_0, \\ \boldsymbol{v}(0) = (v_{x0}, v_{y0}, v_{z0}), \\ \boldsymbol{\omega}(0) = (\omega_{x0}, \omega_{y0}, \omega_{z0}). \end{cases} \tag{7}$$

The observed initial position fixes the origin of the flight. The six velocity and spin components determine its initial motion. In still air, drag removes mechanical energy while Magnus lift does no work along the velocity; gravity converts kinetic and potential energy during flight.

This section presents the equations extended to include buoyancy. The existing numerical results and figures in Section 4 and later sections were calculated using gravity, drag and Magnus lift, corresponding to the baseline $\boldsymbol{F}_B = \mathbf{0}$; the buoyancy-inclusive results have not been recomputed.

4 Model Solving

The following examples retain the existing three-force baseline, without buoyancy. We compare increasing levels of aerodynamic description to identify which effects control the trajectory. Table 2 defines the four models. Forward prediction uses measured or reconstructed launch conditions and unit force corrections. Inverse prediction uses two later positions to estimate the initial motion and force corrections, then propagates the fitted state to the goal.

Table 2: Models and inverse quantities

Model	Assumption	Inverse quantities
M1	gravity	velocity (3)
M2	gravity and drag	velocity, drag coefficient (4)
M3	constant net force	velocity, acceleration (6)
M4	gravity, drag and Magnus force	velocity, spin axis, C_D, α (7)

M3 represents the combined net force, including gravity. M4 uses constant C_D and α with $C_L = \alpha Sp$. Forward reference values are $C_D = 0.2415$ and $\alpha = 1.5251$, anchored to a published reference condition [17].

4.1 Forward Prediction

4.1.1 Prediction procedure and experimental conditions

Forward comparisons use M1, M2 and M4 with prescribed initial conditions and reference aerodynamic coefficients. M3 requires a measured constant acceleration and is therefore excluded from these forward comparisons.

We compare five controlled flights: one nonspinning launch, with a measured range of 21.9 m, and four topspin trajectories. Their initial states are listed in Table 3.

> **Data source and processing.** Goff and Carré [20] used a 32-panel ball (mass 0.424 kg; diameter 0.218 m) and two high-speed cameras, reporting air density 1.2 kg/m^3, kinematic viscosity 1.54×10^{-5} m^2/s and landing-range error at most ± 0.05 m. We extract the red markers in their Figures 3, 4, 11 and 14 from the vector PDF. The nonspinning case uses approximately 18 m/s, 22° and 0.85 m initial height; its early marker spacing is 0.005 s. For the four spinning cases, published Re and Sp determine speed and spin, and the first three extracted positions determine the initial spatial tangent; comparison uses the spatial curve $z(x)$.

Table 3 specifies the forward inputs. All examples lie in the x–z plane, with $y_0 = v_{y0} = 0$ and $\omega_{x0} = \omega_{z0} = 0$; the listed v_{x0}, v_{z0} and ω_{y0} therefore complete the six initial velocity and spin components. Each first marker fixes the initial position.

Table 3: Initial conditions for the controlled forward examples, reconstructed as described above.

Launch	x_0 (m)	z_0 (m)	v_{x0} (m/s)	v_{z0} (m/s)	ω_{y0} (rad/s)
Zero spin	0.0001	0.8501	16.689	6.743	0.00
Topspin a	0.0001	0.8499	16.825	7.366	82.57
Topspin b	0.0001	0.8499	18.194	6.810	44.56
Topspin c	-0.0070	0.8526	17.383	8.188	125.16
Topspin d	-0.0067	0.8500	18.770	10.863	47.75

4.1.2 Calculated trajectories and physical interpretation

Table 4: Nonspinning controlled launch: predicted landing and signed range error. The reported range is 21.9 m.

Model	Range (m)	Prediction minus measurement (m)
M1	24.650	+2.750
M2	20.656	-1.244
M4	20.656	-1.244

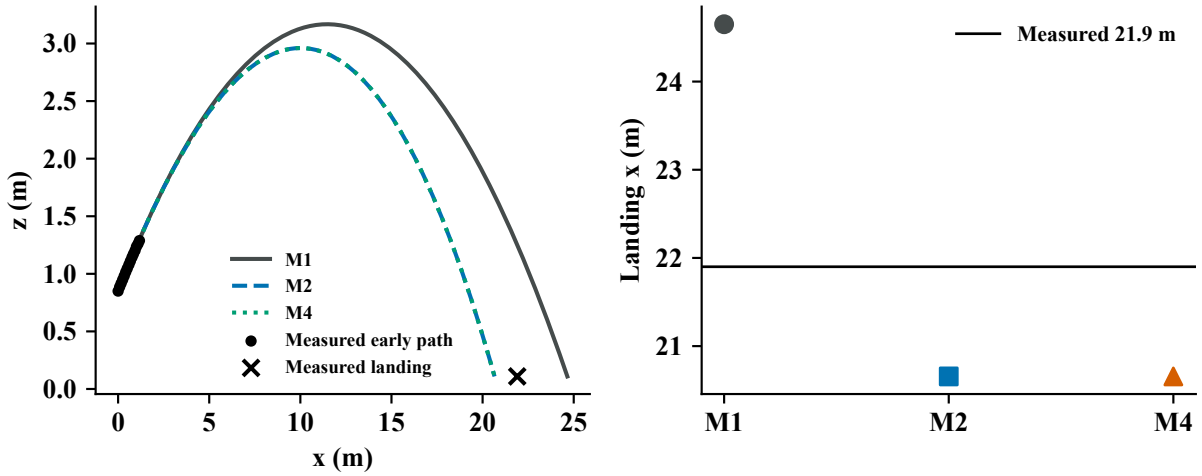


Figure 5: Forward prediction of Goff and Carré's nonspinning experiment [20]. Left: calculated flight paths, digitized early positions and measured landing. Right: individual model ranges against the measured 21.9 m. M2 and M4 coincide because the initial spin is zero.

Drag removes horizontal momentum and shortens flight distance. With zero spin, M2 and M4 use the same reference drag and coincide. The M4 predicted range differs from measurement by -1.244 m.

Table 5: Height RMSE (m) at digitized horizontal positions for the four topspin segments.

Launch	M1	M2	M4
a	0.0119	0.0118	0.0036
b	0.0396	0.0397	0.0480
c	0.0100	0.0100	0.0139
d	0.0171	0.0170	0.0069

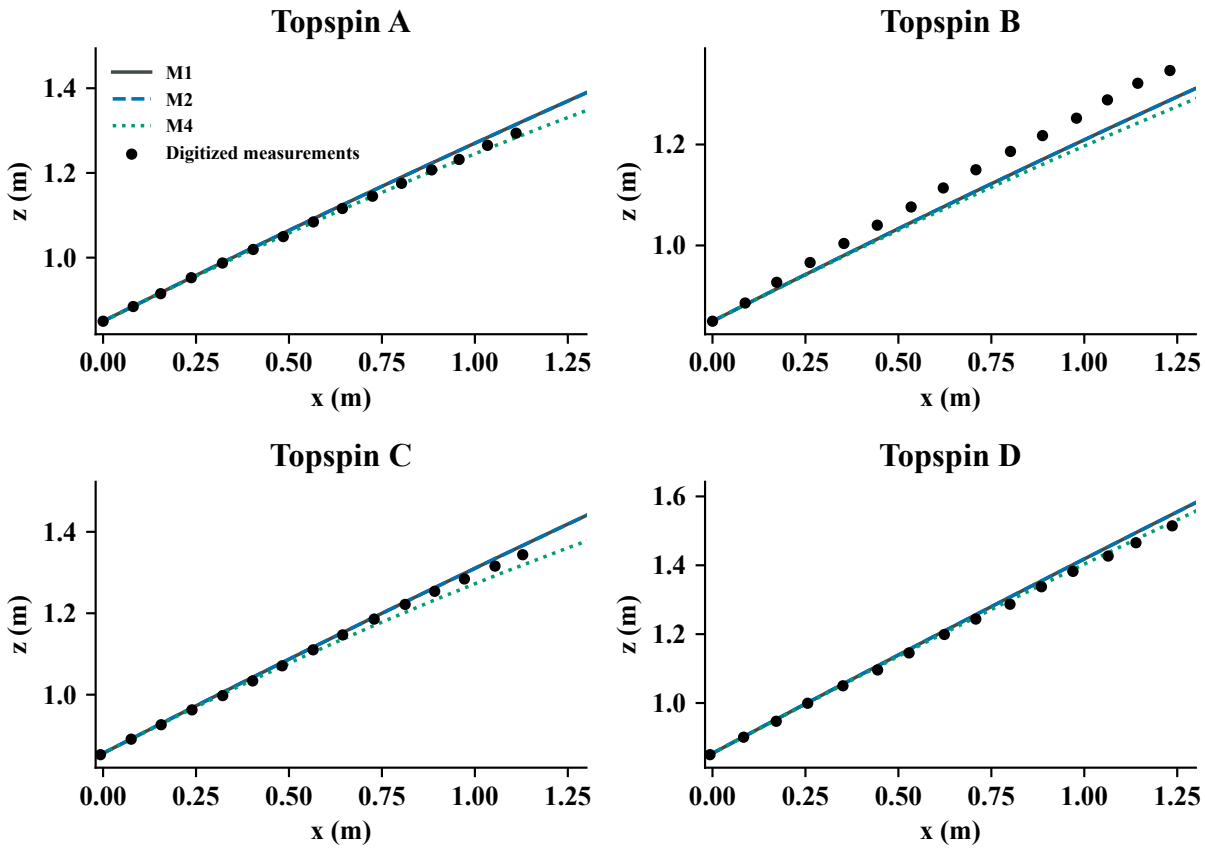


Figure 6: Four topspin launches digitized from Goff and Carré [20], Figure 14. Black points are measured marker centres; model curves use unit force corrections and the stated initial conditions. Each panel covers the published early spatial segment.

Topspin produces downward Magnus force. Table 5 reports height RMSE at the measured horizontal positions.

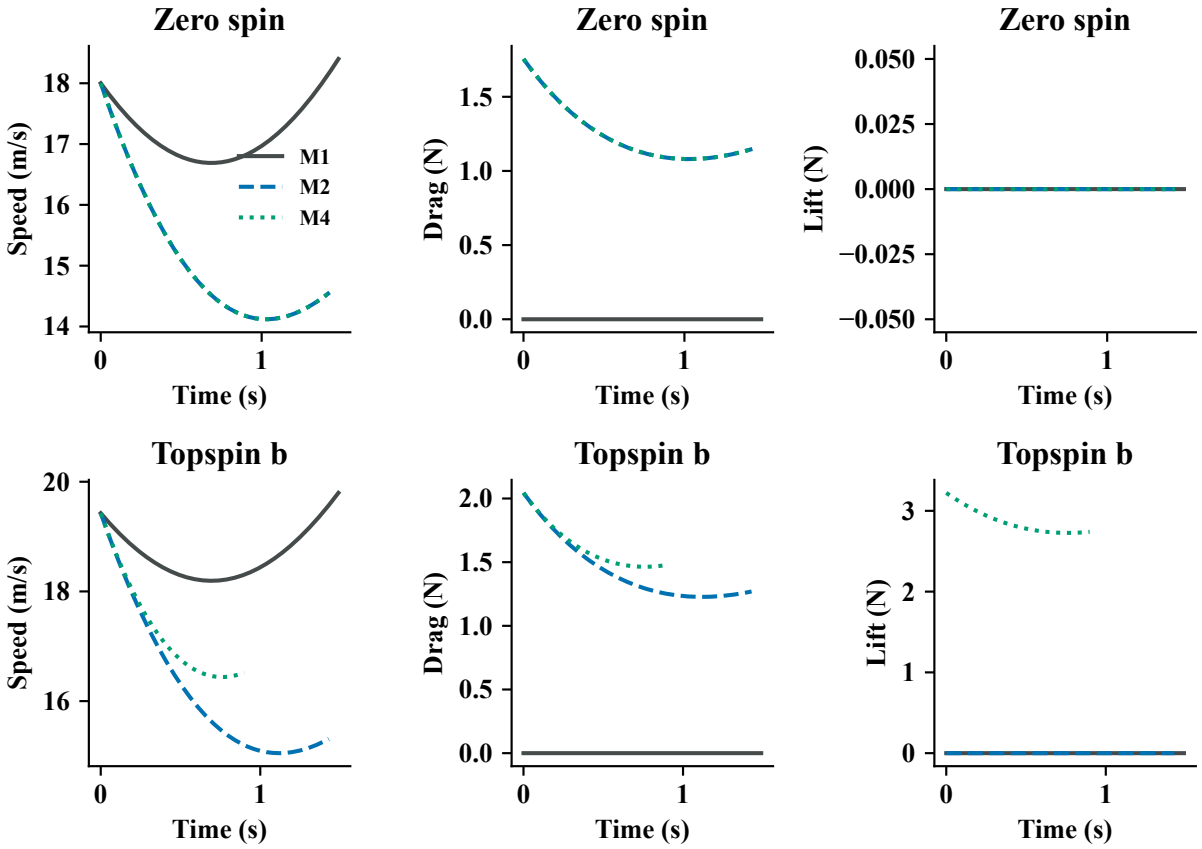


Figure 7: Speed, drag and lift during the nonspinning and topspin-b forward examples. Force magnitudes are calculated from the shared equations. Constant aerodynamic coefficients still produce changing forces because dynamic pressure varies with speed.

Drag is strongest when the ball is fast. Downward lift in a topspin flight changes its vertical motion and flight duration. With fixed coefficients, force magnitude still changes with speed.

4.2 Inverse Prediction

4.2.1 Early-position fitting and representative solution

M3 solves for constant acceleration from three positions:

$$\boldsymbol{r}(t) = \boldsymbol{r}_0 + \boldsymbol{v}_0 t + \tfrac{1}{2}\boldsymbol{a}t^2. \tag{8}$$

M4 minimizes the two later position errors over launch velocity, spin direction, C_D and α. To obtain a representative solution, the spin magnitude is fixed at 30 rad/s; multiple deterministic initial guesses are used within the specified bounds. Only the product $\alpha\boldsymbol{\omega}$ appears in Magnus force, so position data do not separately identify α and spin magnitude. Seven unknowns are constrained by six later position coordinates; the result is a representative bounded solution.

In this inversion, C_D and α are each restricted to 0.5–1.5 times their reference values. Both reach their upper bounds in the competition case, so the representative solution depends on the

selected bounds; a wider search and comparison of feasible solutions are needed to assess stability.

4.2.2 Comparison with recorded match flights

The Sportec Open DFL dataset associates optical tracking with match events [10, 11]. We analyze four flights from matches J03WOH and J03WQQ, and one recorded penalty. Events are selected for an identifiable shot and a tracked crossing of the actual goal-line plane. The four flights comprise two headers and two foot shots. Each inverse calculation uses the first three available positions at 0, 0.04 and 0.08 s. Later tracking coordinates determine the observed crossing of the goal plane. The penalty segment begins with a slower first displacement before the faster recorded flight; its early coordinates are used as recorded.

Table 6: All five recorded flights and all four inverse models. Error is the Euclidean difference in goal-plane (y, z).

Event	Model	Termination	Early residual (m)	Plane error (m)
WOH 0487 (head)	M1	plane	0.036	0.717
WOH 0487 (head)	M2	plane	0.0317	0.700
WOH 0487 (head)	M3	ground	0	—
WOH 0487 (head)	M4	plane	0.0313	1.392
WOH 0606 (foot)	M1	plane	0.0263	4.988
WOH 0606 (foot)	M2	ground	0.0283	—
WOH 0606 (foot)	M3	time limit	0	—
WOH 0606 (foot)	M4	plane	0.0228	8.815
WOH 0968 (foot)	M1	ground	0.124	—
WOH 0968 (foot)	M2	ground	0.125	—
WOH 0968 (foot)	M3	plane	0	1.083
WOH 0968 (foot)	M4	ground	0.121	—
WQQ 1227 (head)	M1	plane	0.18	2.785
WQQ 1227 (head)	M2	plane	0.178	2.860
WQQ 1227 (head)	M3	plane	0	13.528
WQQ 1227 (head)	M4	plane	0.174	3.700
Penalty	M1	plane	0.204	0.853
Penalty	M2	plane	0.206	0.856
Penalty	M3	plane	0	0.968
Penalty	M4	plane	0.207	0.207

Table 7: Reached-case errors and termination counts. Individual errors are retained in the preceding table.

Model	Reached	Ground	Other	Fit failures	Mean error (m)
M1	4/5	1	0	0	2.336
M2	3/5	2	0	0	1.472
M3	3/5	1	1	0	5.193
M4	4/5	1	0	0	3.529

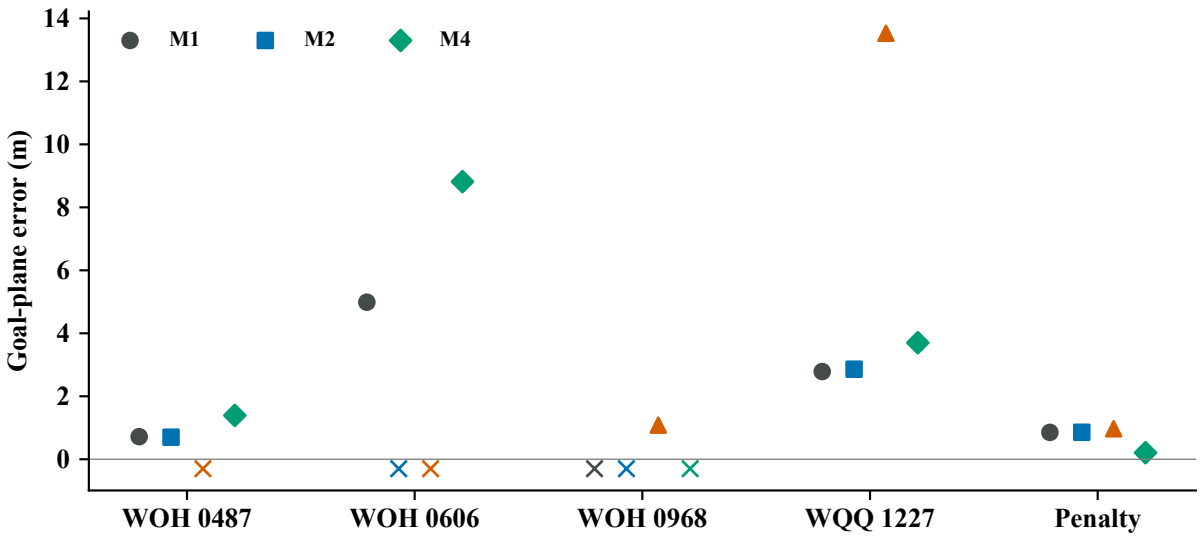


Figure 8: Individual goal-plane prediction errors for the five recorded flights. Each marker is one event and model. Crosses below zero denote trajectories that provide no goal-plane prediction; their termination status appears in Table 6.

M1 reaches the goal plane in 4 of five cases; M2 reaches the goal plane in 3 of five cases; M3 reaches the goal plane in 3 of five cases; M4 reaches the goal plane in 4 of five cases. M1 gives the largest number of goal-plane predictions in these five recorded flights. The table reports each reached-case mean together with its contributing count. Large force corrections and iteration-limit cases reveal a strong dependence on the early recorded motion. Fit failures denote numerical iteration limits; their candidate trajectories are retained in the calculation records.

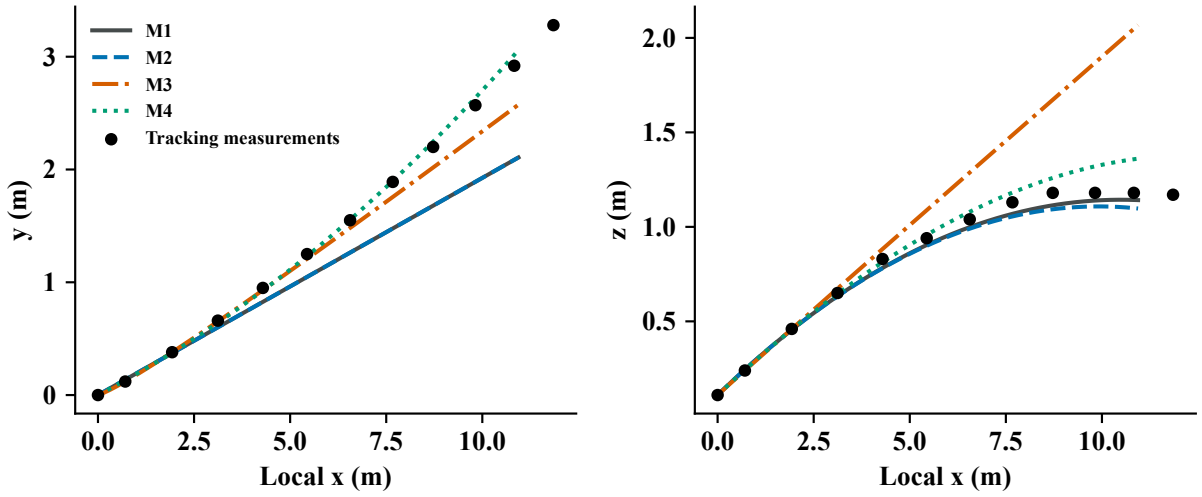


Figure 9: Converged inverse predictions for the recorded penalty, with all tracked positions shown. The first three positions enter the fit; remaining positions display the observed continuation. All model statuses are listed in Table 6.

The comparison links early acceleration to later flight. A large fitted drag correction produces rapid momentum loss, while lift corrections alter the direction of curvature. Changes in early recorded displacement therefore have a strong effect on the predicted endpoint. The event-level results show how each model responds to the actual recorded segment.

We also recomputed the three penalty flights after the teammate corrected their starting frames. The table reports goal-plane errors in metres; a dash denotes ground contact, early termination or an unconverged fit.

Table 8: Corrected-start penalties: goal-plane error (m)

Case	M1	M2	M3	M4
J03WMX	—	—	—	0.360
J03WOH	0.684	0.657	0.374	0.113
J03WR9	0.550	0.541	—	—

4.2.3 Competition Problem B

The supplied observations are

t (s)	x (m)	y (m)	z (m)
0	0	0	0.11
0.05	1.50	0.45	0.44
0.10	2.96	0.86	0.74

(9)

Applying the same inverse procedure gives Table 9. The preferred M4 initial conditions are

$$\begin{aligned} \boldsymbol{v}_0 &= (30.378, 9.407, 6.897)\ \mathrm{m/s}, \\ \boldsymbol{\omega}_0 &= (19.743, 1.206, -22.555)\ \mathrm{rad/s}, \\ (C_D, \alpha) &= (0.3623, 2.2877). \end{aligned} \tag{10}$$

Table 9: Inverse predictions for the competition observations. Residual is the maximum absolute early-coordinate difference.

Model	C_D	α	Residual (m)	y_g (m)	z_g (m)	t_g (s)
M1	—	—	0.016	3.217	1.957	0.3706
M2	—	—	0.011	3.217	1.883	0.4041
M3	—	—	0	2.495	1.920	0.4050
M4	0.3623	2.2877	0.00078	2.616	1.953	0.3990

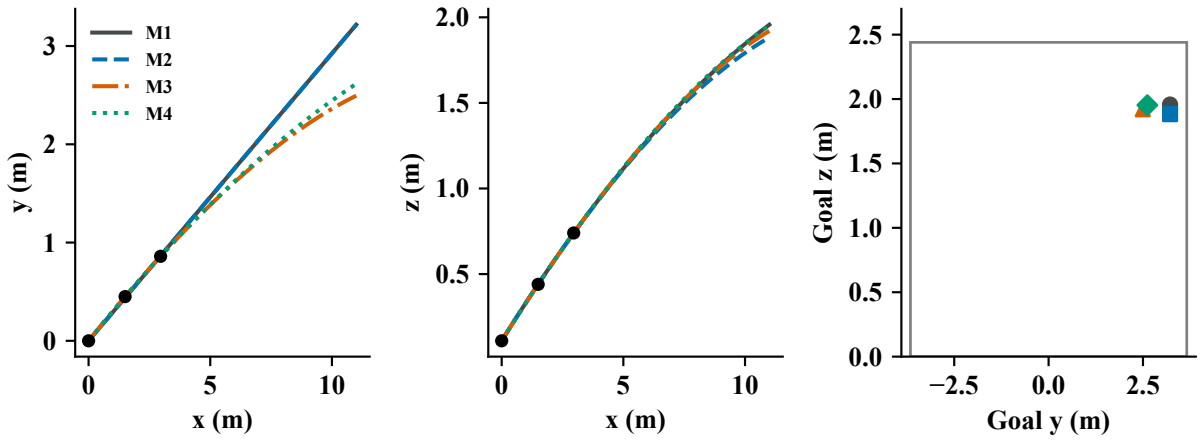


Figure 10: Four inverse predictions for Problem B: horizontal path, vertical path and ball-centre positions at the goal plane. Black points are the supplied early observations.

M1 and M2 predict nearly the same horizontal direction. M3 extends the constant acceleration determined from three positions; M4 describes curvature through Magnus force. The M4 goal-plane centre is $(2.616, 1.953)$ m at 0.3990 s, with early maximum coordinate residual 0.00078 m. The stated spin magnitude is a reference setting.

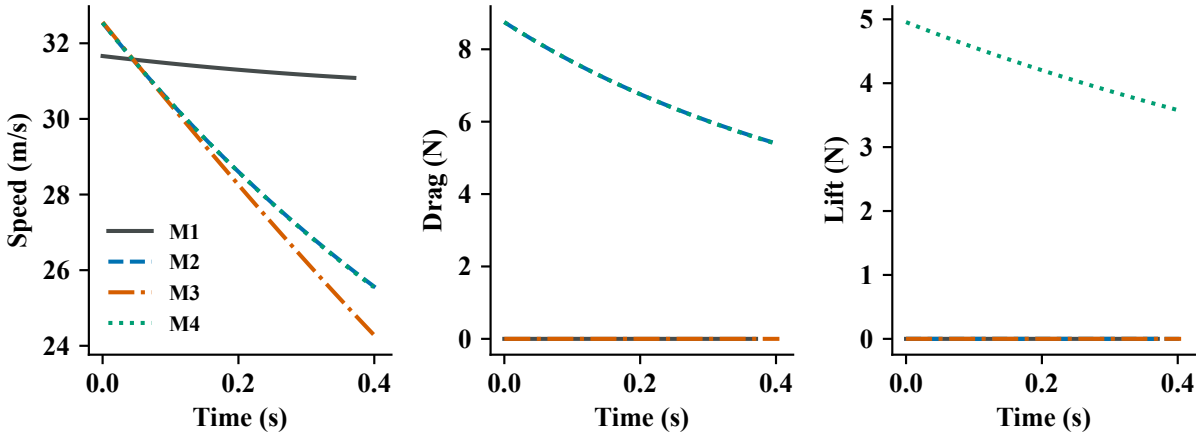


Figure 11: Speed and aerodynamic force magnitudes for the four fitted Problem B models. All share the same observations, while their force assumptions produce different continuation paths.

4.3 Case Studies: Different Penalty-Kick Techniques

What happens when a goalkeeper faces not one textbook kick but a repertoire of different techniques? We examine five illustrative situations using the same three early positions at 0, 0.05 and 0.10 s. Each reference flight is generated with M4; its first three positions feed M1 and M4 with zero assumed spin. The figures show the generated flight, the zero-spin M4 continuation and

the input points. The chosen values describe examples, not statistical means or individual players [6, 9, 7, 8]. No observation noise or wind is added.

Case 1: Power Shot

The ball leaves at 27 m/s with an 8° elevation and zero spin. It reaches the goal quickly; in this example the M1 goal-plane error is 0.060 m, and the zero-spin M4 result is within 0.001 m. A goalkeeper has little time to react, although the weakly curved path is relatively straightforward under these assumed conditions.

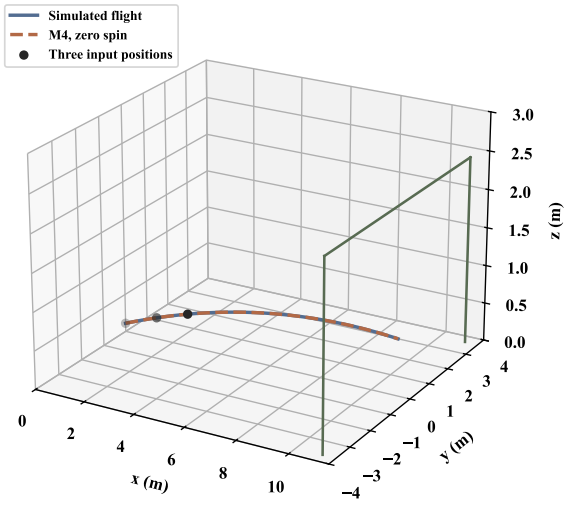


Figure 12: Case 1: power shot.

Case 2: Placement Shot

For a more controlled shot, we choose 20 m/s, an 11° elevation and zero spin. The simulated ball crosses the goal plane at a height of 0.551 m. M1 has a 0.123 m error, while zero-spin M4 is within 0.001 m. A lower speed leaves more flight time, so the goalkeeper can use the developing path to refine the expected position.

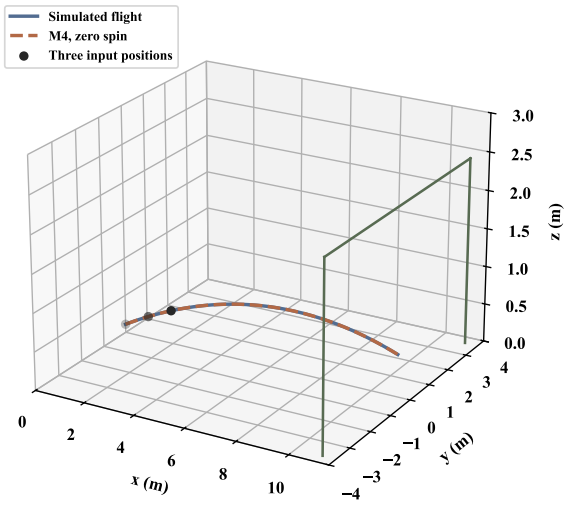


Figure 13: Case 2: placement shot.

Case 3: Curved Shot

The curved kick uses 24 m/s, a 10° elevation and vertical-axis spin $(0, 0, 35)$ rad/s. The flight bends sideways: its simulated goal crossing is $y = 0.841$ m, $z = 0.877$ m. When the prediction assumes zero spin, M4 misses this crossing by 0.688 m. The first positions alone can make the sideways motion appear partly attributable to initial direction, so a goalkeeper should allow for subsequent curvature.

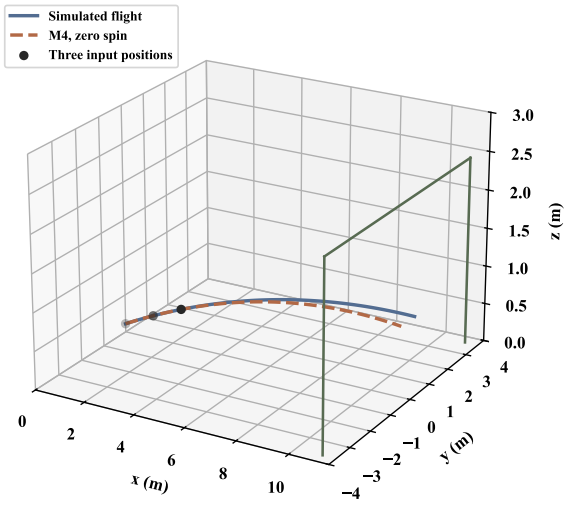


Figure 14: Case 3: sidespin curve.

Case 4: Topspin and Backspin

Both examples use 24 m/s. The topspin case takes 12° and $(0, 35, 0)$ rad/s; the backspin case takes 10° and $(0, -35, 0)$ rad/s. Their simulated goal heights are 0.426 m and 1.723 m, respectively. Zero-

spin M4 incurs errors of 0.682 m and 0.692 m. Opposite spin changes the vertical path, making height an important part of the goalkeeper's assessment.

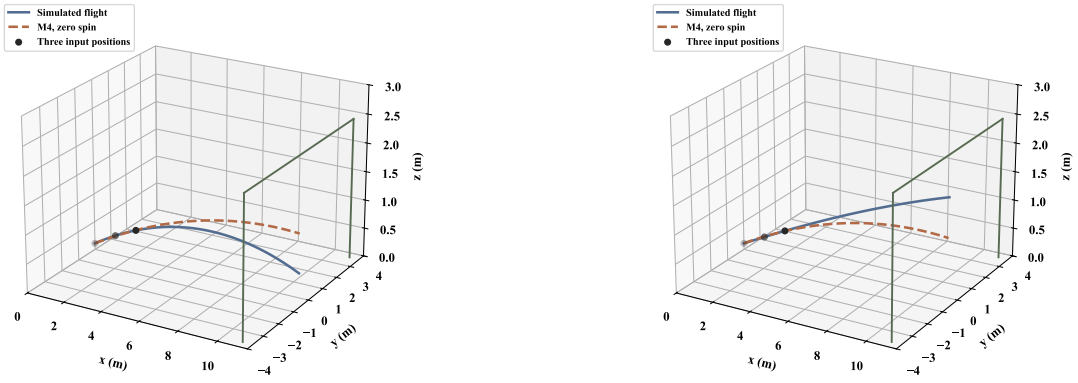


Figure 15: Case 4: topspin (left) and backspin (right).

Case 5: Chip Shot

The chip uses 14 m/s, a 20° elevation and $(0, -15, 0)$ rad/s backspin. Its higher launch angle and lower speed produce a longer, arcing flight; the simulated goal height is 0.941 m. M1 and zero-spin M4 have errors of 0.250 m and 0.555 m, respectively. Unlike a power kick, the goalkeeper must account for a slower ball that rises before descending.

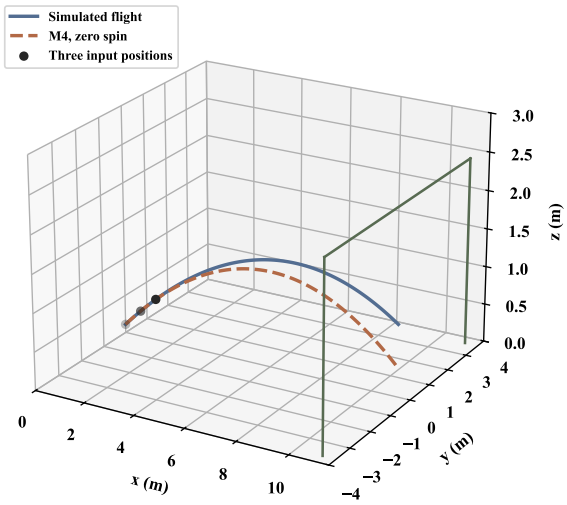


Figure 16: Case 5: chip shot.

Comparison. Table 10 collects the six trajectories. M4 with the generating spin supplied is a conditional control, not a three-position-only prediction: its near-zero error follows from using the same equations to generate and predict the trajectory. These single synthetic examples illustrate possible effects of spin; the recorded-match comparison remains a separate evaluation. These cases use the same fixed M4 coefficients.

Table 10: Summary of illustrative three-point cases; goal-plane errors in metres

Case	v_0 (m/s)	Angle (deg)	Spin (ω_y, ω_z) (rad/s)	M1 (m)	M4 zero (m)	M4 known (m)
Power	27	8	$(0, 0)$	0.060	< 0.001	< 0.001
Placement	20	11	$(0, 0)$	0.123	< 0.001	< 0.001
Curved	24	10	$(0, 35)$	0.693	0.688	< 0.001
Topspin	24	12	$(35, 0)$	0.764	0.682	< 0.001
Backspin	24	10	$(-35, 0)$	0.610	0.692	< 0.001
Chip	14	20	$(-15, 0)$	0.250	0.555	< 0.001

5 Sensitivity Analysis

To be completed.

6 Discussion of Other Factors

We now examine surface geometry, spin evolution and ambient airflow.

6.1 Asymmetric Surface Geometry

Paudel and Yan [14] performed full-flight flow simulations using realistic football geometry and a smooth-sphere comparison. For launch speeds of 10, 15 and 20 m/s and spin rates of 31.5, 63 and 94.5 rad/s, their results show that seams and surface structure alter separation, wake dynamics and Magnus force. The smooth sphere produces greater lateral bending in their nine comparisons, and vertical flight-time differences become more visible at higher speed and rotation.



Figure 17: Illustrative comparison of a paneled football and a smooth sphere, inspired by Paudel and Yan [14]. Their compared bodies had diameter 0.22 m and mass 0.436 kg. The panel geometry shown here is schematic.

The wind-tunnel measurements of Alam et al. [13] show that football surface structure affects the drag coefficient.

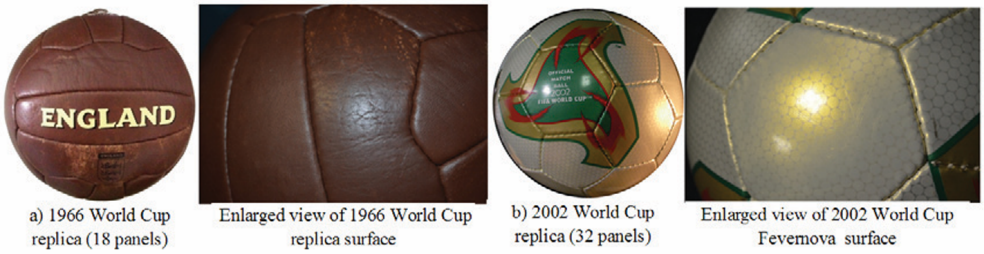


Fig. 1. Vintage and Adidas Fevernova balls



Fig. 2. Adidas Teamgeist 2006 and 2009 balls



Fig. 3. Adidas 2010 Jabulani and Nike 2010 T-90 balls

Figure 18: Six footballs and details of their surfaces, reproduced from Figures 1–3 of Alam et al. [13].

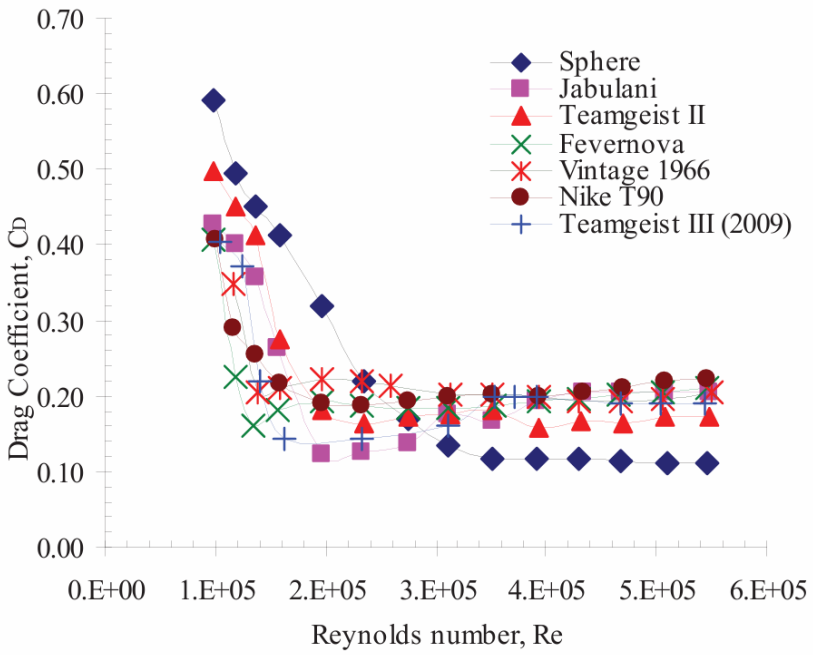


Figure 19: Drag coefficient against Reynolds number for different footballs and a smooth sphere, reproduced from Figure 5 of Alam et al. [13].

The spherical geometry in the present report describes the centre-of-mass motion and projected area. Its force coefficients come from football measurements, so the average surface influence enters through C_D and C_L. Explicit panel orientation and asymmetric wake fluctuations provide a further level of description when lateral-force details are important. These published results support retaining football-specific coefficients in a spherical trajectory model.

6.2 Rotation and Aerodynamic Damping

Aerodynamic torque can reduce spin [18, 19]; the main model holds spin constant. The linear-spin lift expression can support subsequent comparisons of spin magnitude and damping.

6.3 Wind Speed and Force Estimates

Wind changes the airflow through $\boldsymbol{u} = \boldsymbol{v} - \boldsymbol{v}_{\text{wind}}$ [12]. At fixed coefficients, a wind speed w along the flight direction changes the dynamic-pressure scale by

$$\frac{q_{\text{head}}}{q_0} = (1 + w/v)^2, \qquad \frac{q_{\text{tail}}}{q_0} = (1 - w/v)^2, \qquad \frac{q_{\text{cross}}}{q_0} = 1 + (w/v)^2. \tag{11}$$

Crosswind also rotates the airflow by $\arctan(w/v)$, changing the force direction. Table 11 evaluates these rough estimates at the fitted initial speed. The listed wind speeds are illustrative conditions for comparison.

Table 11: Illustrative wind estimates at the fitted launch speed. Percentages describe dynamic pressure at fixed aerodynamic coefficients.

w (m/s)	Headwind (%)	Tailwind (%)	Crosswind (%)	Airflow angle (°)
1	+6.24	-6.05	+0.09	1.76
3	+19.29	-17.59	+0.85	5.27
5	+33.09	-28.37	+2.36	8.74

A 5% limit on the headwind dynamic-pressure change corresponds to $w/v \leq \sqrt{1.05} - 1$, or $w \leq 0.80$ m/s at this launch. The still-air approximation is most suitable for indoor or weak-wind flight; stronger wind should enter through the relative-velocity calculation.

7 Conclusions

The model relates soccer-ball flight to gravity, buoyancy, drag and Magnus lift; the numerical results presented here use the three-force baseline without buoyancy. M4 uses constant drag and linear-spin lift to reduce the number of fitted parameters.

The controlled nonspinning example shows that including drag is essential for range prediction. Its M4 range is 20.656 m against the published 21.9 m. The rotating trajectory segments display the downward curvature of topspin and variation in the accuracy of the common coefficient law. Recorded match comparisons show how early displacement and fitted force corrections affect subsequent prediction.

For Problem B, M4 gives $(y, z) = (2.616, 1.953)$ m at the goal plane. Magnus lift accounts for the marked horizontal difference from the gravity-only and drag-only predictions. Wind estimates give the weak-wind condition for the still-air approximation. Sensitivity-analysis findings will be added after Section 5 is completed.

8 Model Evaluation

8.1 Physical Interpretation and Computational Use

The hierarchy makes the contribution of each force visible. The same dynamics handles forward calculation, early-position inversion and endpoint prediction. Force corrections provide a direct way to accommodate measured trajectory differences, and all experimental and numerical inputs are retained with their sources.

8.2 Predictive Performance and Improvements

The wind-tunnel comparisons, controlled launches and recorded match flights examine complementary aspects of the model: coefficient trends, curvature and range, and prediction from early observations. Their event-level errors identify where a common football coefficient law and an early trajectory segment give a useful continuation. Ball-specific aerodynamic measurements, directly measured spin and synchronized higher-rate positions would improve this connection. Ex-

plicit surface orientation, rotational damping and wind can be included as the application requires.

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Reproduction Code

The accompanying project contains the frozen inputs, vector-marker extraction, four-model calculation, result tables and figure scripts. The following excerpts show the shared state equation and force evaluation. The inverse solver uses the same state derivative with the three selection criteria described in Section 4.2. Calculation and figure generation are separate commands, allowing plots to be refreshed from saved results.

```
--
    return -factor*w


def state_derivative(t, state, config):
    """The single nine-state RHS used for prediction and inverse fitting."""
```

```
--
    return record


def forces(velocity, spin, config):
    """Return force components on ball, in newtons."""
    v = np.asarray(velocity, dtype=float) - np.asarray(config.wind, dtype=float)
    omega = np.asarray(spin, dtype=float)
    speed = np.linalg.norm(v)
    weight = (config.mass * np.asarray(config.constant_acceleration, dtype=float)
              if config.constant_acceleration is not None else
              np.array([0., 0., -config.mass * config.gravity if config.use_gravity else
                  0.]))
    drag = np.zeros(3)
    magnus = np.zeros(3)
    if speed > 1e-12:
        cd, cl, _, _, _ = coefficients(velocity, spin, config)
        q_area = 0.5 * config.air_density * np.pi * config.radius**2 * speed**2
        if config.use_drag:
            drag = -q_area * config.drag_scale * cd * v / speed
        cross = np.cross(omega, v)
        norm = np.linalg.norm(cross)
        if config.use_magnus and norm > 1e-12:
```